matlab.mathworks.com

MATLAB Drive | MATLAB

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FILE | NAVIGATE | CODE | ANALYZE | SECTION | RUN

/ › MATLAB Drive

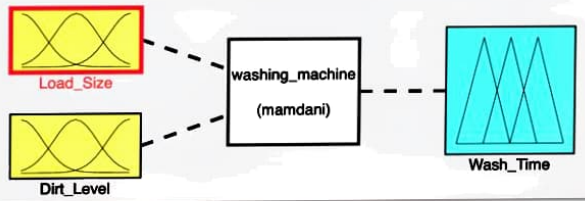
washing_machine.fis

/MATLAB Drive/washing_machine.fis

```
[System]
Name='washing_machine'
Type='mamdani'
Version=2.0
NumInputs=2
NumOutputs=1
NumRules=6
AndMethod='min'
OrMethod='max'
ImpMethod='min'
AggMethod='max'
DefuzzMethod='centroid'

[Input1]
Name='Load_Size'
Range=[0 10]
NumMFs=3
MF1='Low':'trimf',[0 0 5]
MF2='Medium':'trimf',[2 5 8]
MF3='High':'trimf',[5 10 10]

[Input2]
Name='Dirt_Level'
Range=[0 10]
NumMFs=3
MF1='Low':'trimf',[0 0 5]
MF2='Medium':'trimf',[2 5 8]
MF3='High':'trimf',[5 10 10]

[Output1]
Name='Wash_Time'
Range=[0 60]
NumMFs=3
MF1='Short':'trimf',[0 0 20]
MF2='Medium':'trimf',[15 30 45]
MF3='Long':'trimf',[40 60 60]
```

Fuzzy Logic Designer: washing_machine

File Edit View



FIS Name: washing_machine FIS Type: mamdani

And method	min
Or method	max
Implication	min
Aggregation	max
Defuzzification	centroid

Current Variable

Name	Load_Size
Type	input
Range	[0 10]

Help | Close

Saved FIS "washing_machine" to file

Command Window

>> Press ⇧⌘P to generate code with Copilot

Copilot Chat

Copilot

Hello! Copilot is here to answer your questions, help write and explain code, and even identify code issues. Learn more about Copilot.

Copilot sometimes produces output that seems accurate but is not. Validate generated output before use. Share feedback on the output to help improve the responses.

Shuffle Example Prompts

- Solve the linear equations with coefficients A = [2, 4; 1, 3] and constants B = [8; 5] and display the results
- Create a 3D bar graph of data from magic function
- Generate noisy data from a cubic polynomial, preprocess the data, and plot the results

Ask Copilot

Validate generated output before use.

Editor: 100% UTF-8 LF plain text file Ln 1 Col 1

Rule Editor: Untitled

MATLAB Online

File Edit View Options

1. If (Load_Size is Low) and (Dirt_Level is Low) then (Wash_Time is Short) (1)
2. If (Load_Size is Medium) and (Dirt_Level is Low) then (Wash_Time is Short) (1)
3. If (Load_Size is Medium) and (Dirt_Level is Medium) then (Wash_Time is Medium) (1)
4. If (Load_Size is Low) and (Dirt_Level is High) then (Wash_Time is Medium) (1)
5. If (Load_Size is High) and (Dirt_Level is High) then (Wash_Time is Long) (1)
6. If (Load_Size is High) and (Dirt_Level is Medium) then (Wash_Time is Long) (1)

If Load_Size is	and Dirt_Level is	Then Wash_Time is
Low	Low	Short
Medium	Medium	Medium
High	High	Long
none	none	none
☐ not	☐ not	☐ not

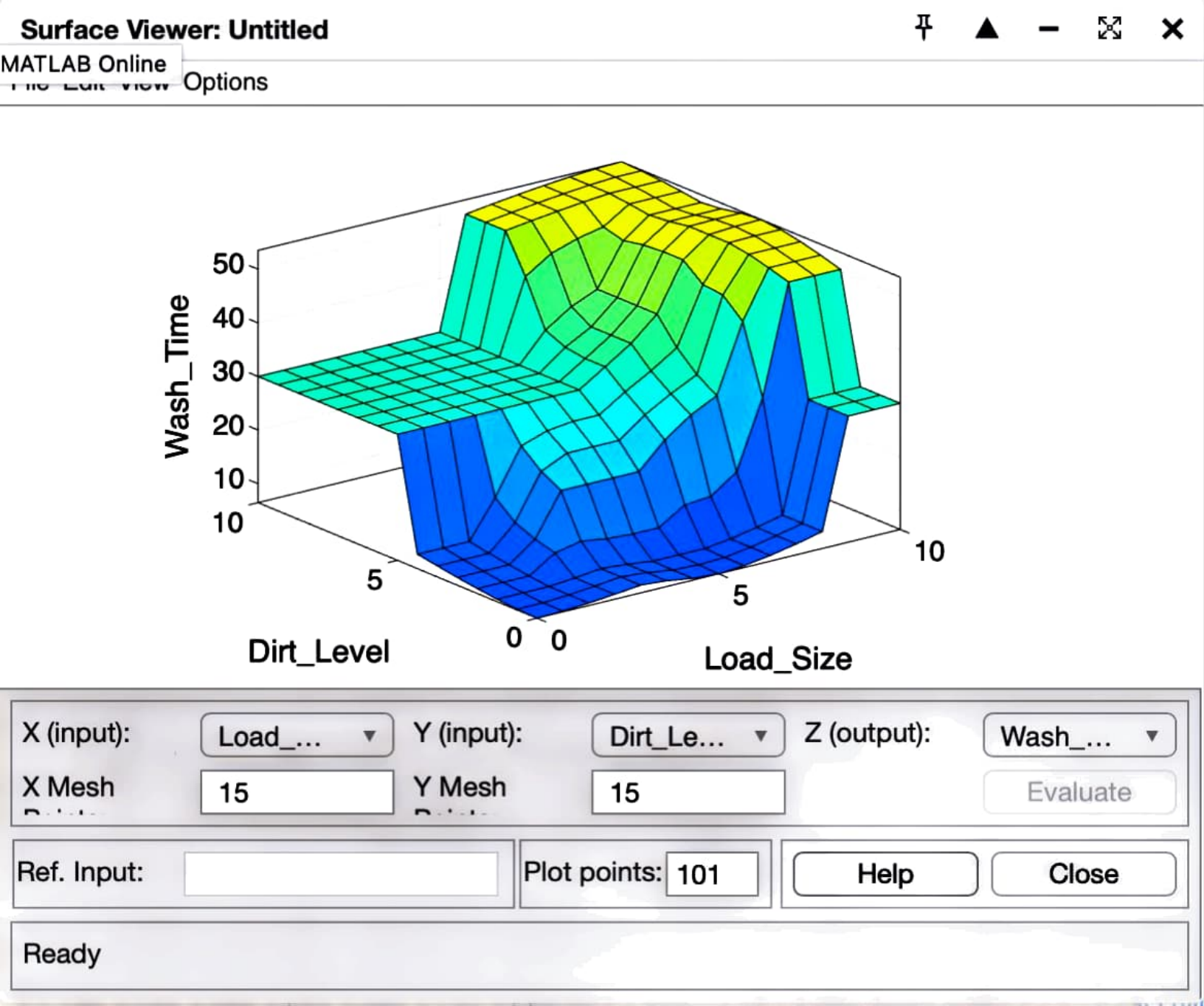
Connection
- ○ or
- ◉ and

Weight: 1

Delete rule | Add rule | Change rule | << | >>

The rule is added

Help | Close


Surface Viewer: Untitled
MATLAB Online
Options
Wash_Time
50
40
30
20
10
10
5
0
Dirt_Level
0
5
10
Load_Size
X (input):
Load_...
Y (input):
Dirt_Le...
Z (output):
Wash_...
X Mesh
15
Y Mesh
15
Evaluate
Ref. Input:
Plot points:
101
Help
Close
Ready

Membership Function Editor: Untitled

MATLAB Online

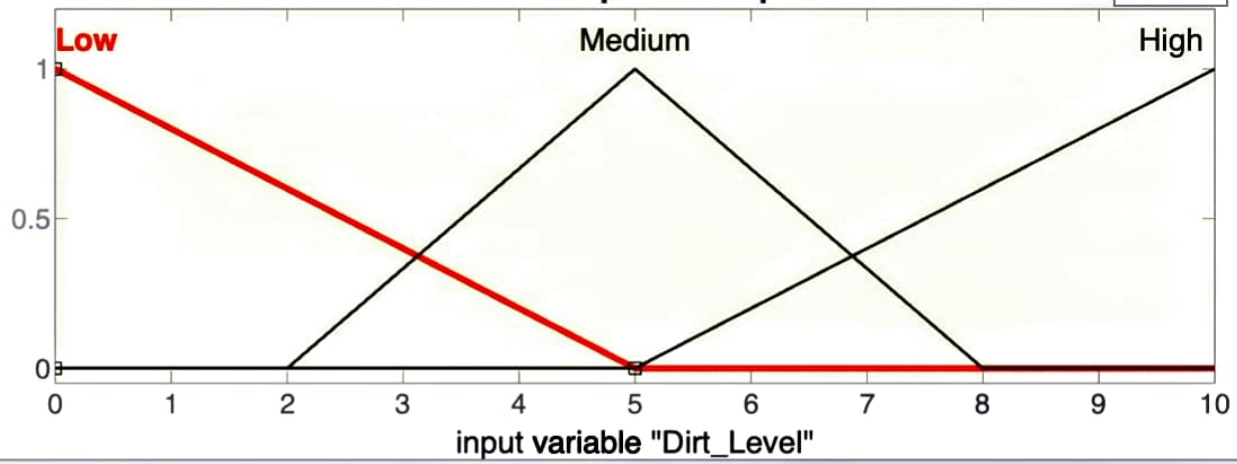
FIS Variables

Load_Size Wash_Time

Dirt_Level

Membership function plots

plot points: 181

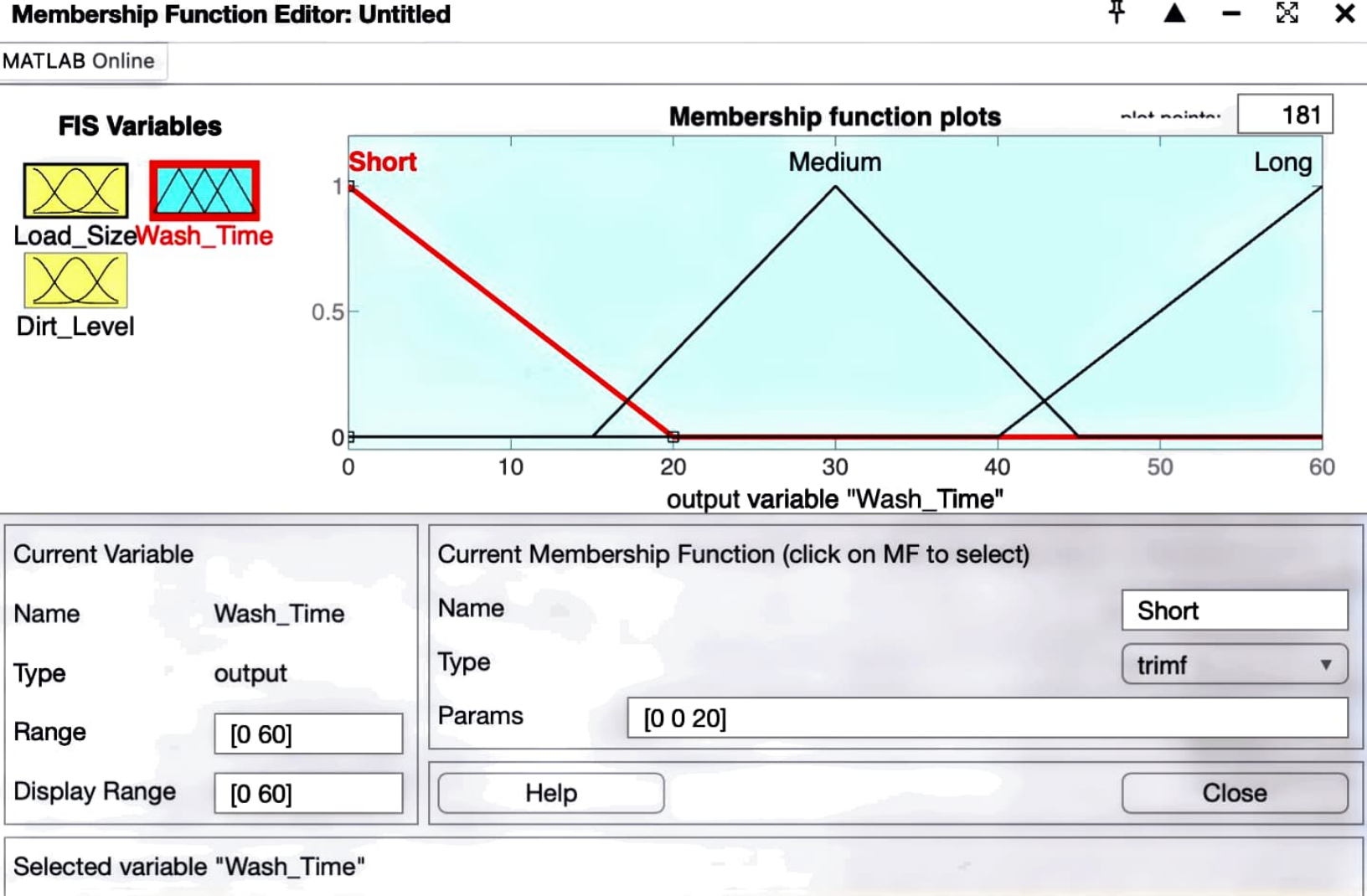


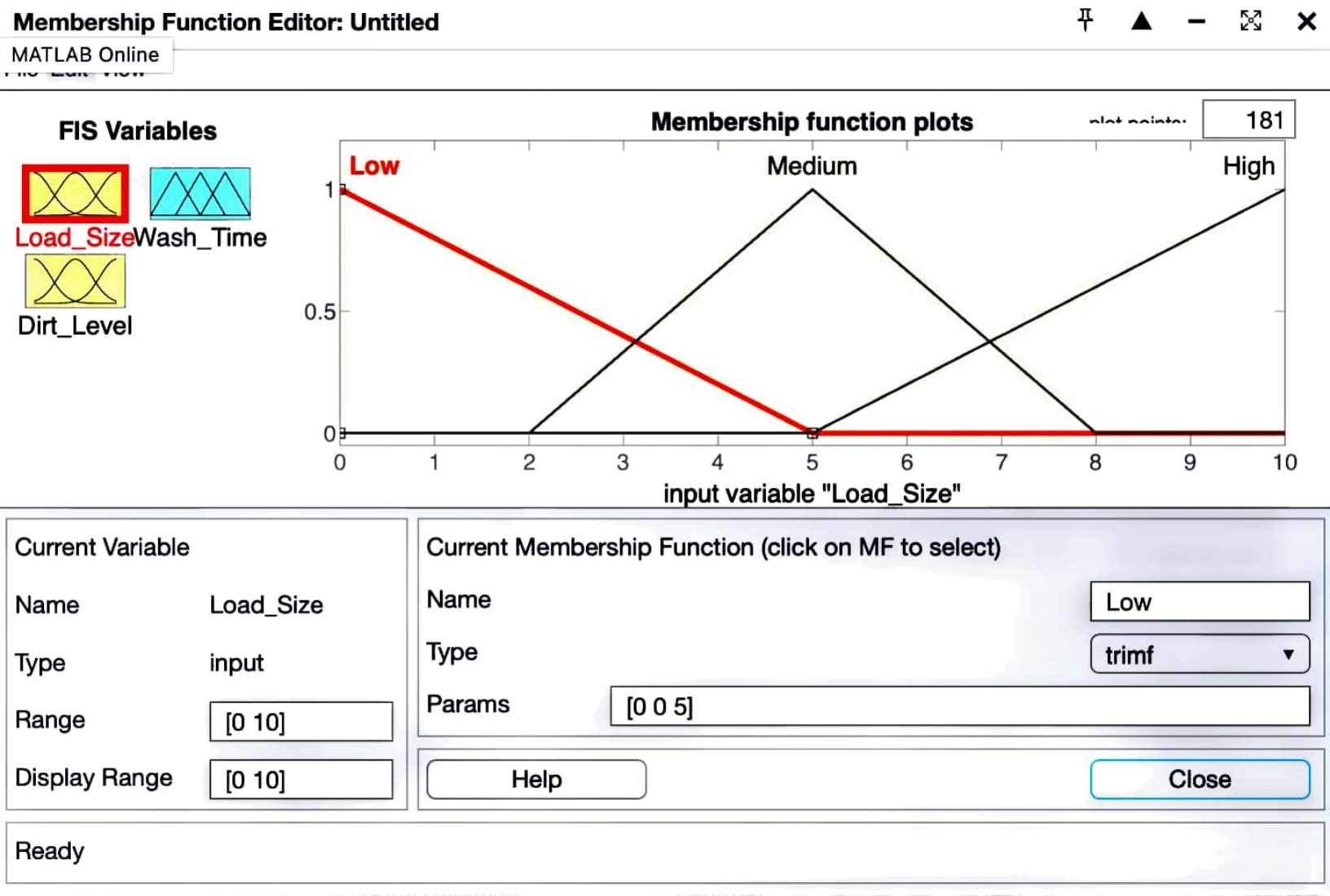
Current Variable	
Name	Dirt_Level
Type	input
Range	[0 10]
Display Range	[0 10]

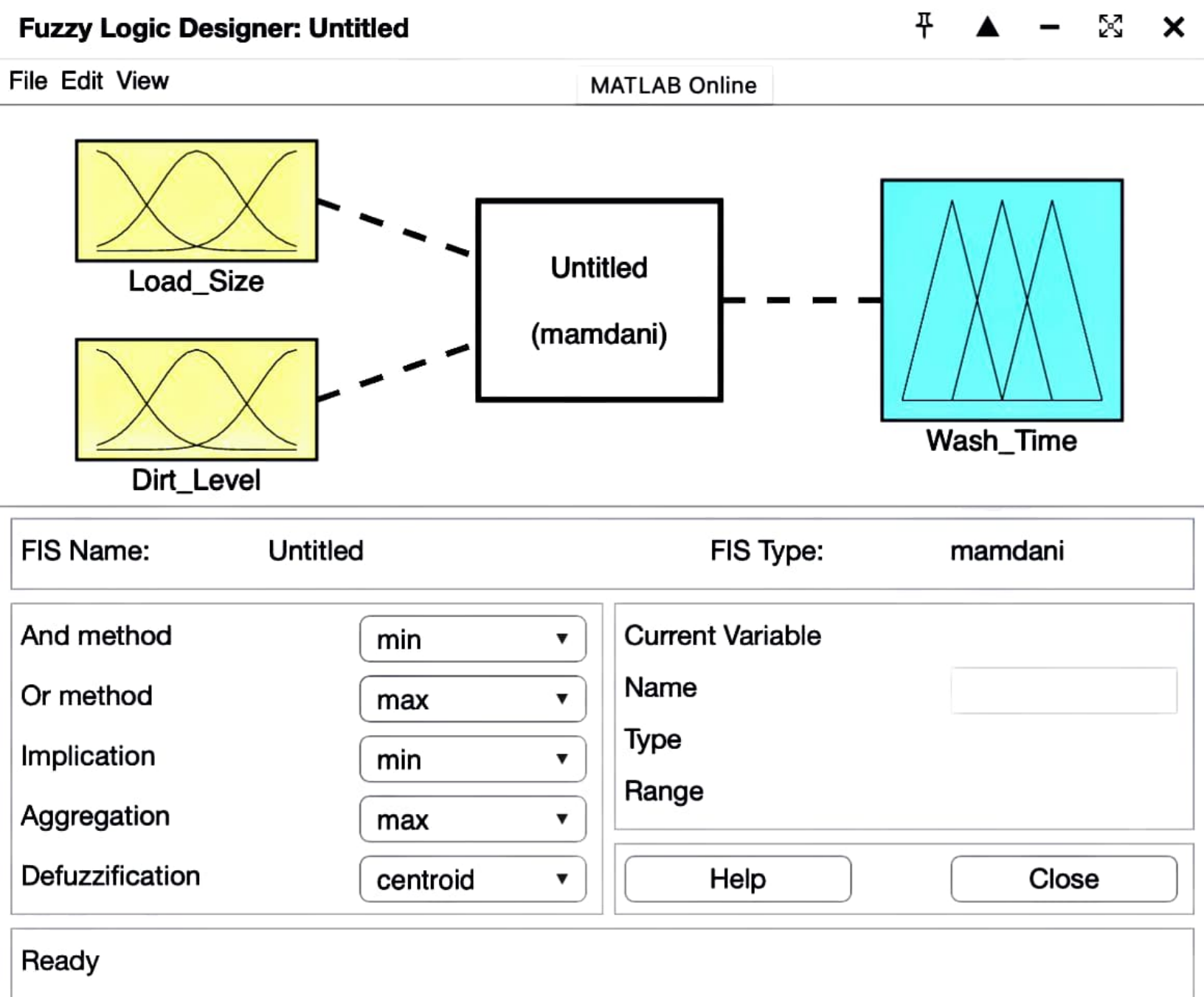
Current Membership Function (click on MF to select)	
Name	Low
Type	trimf
Params	[0 0 5]

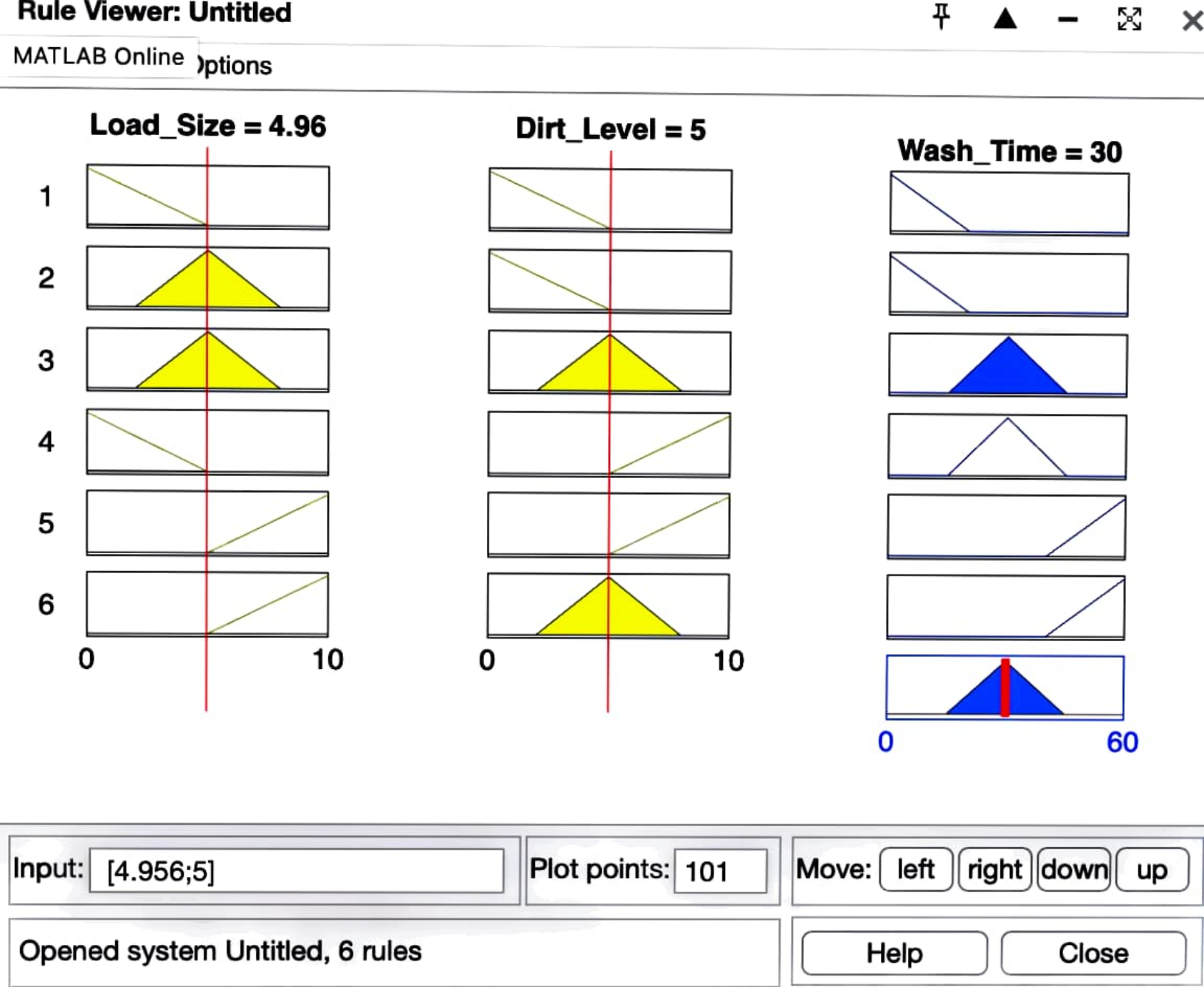
Help Close

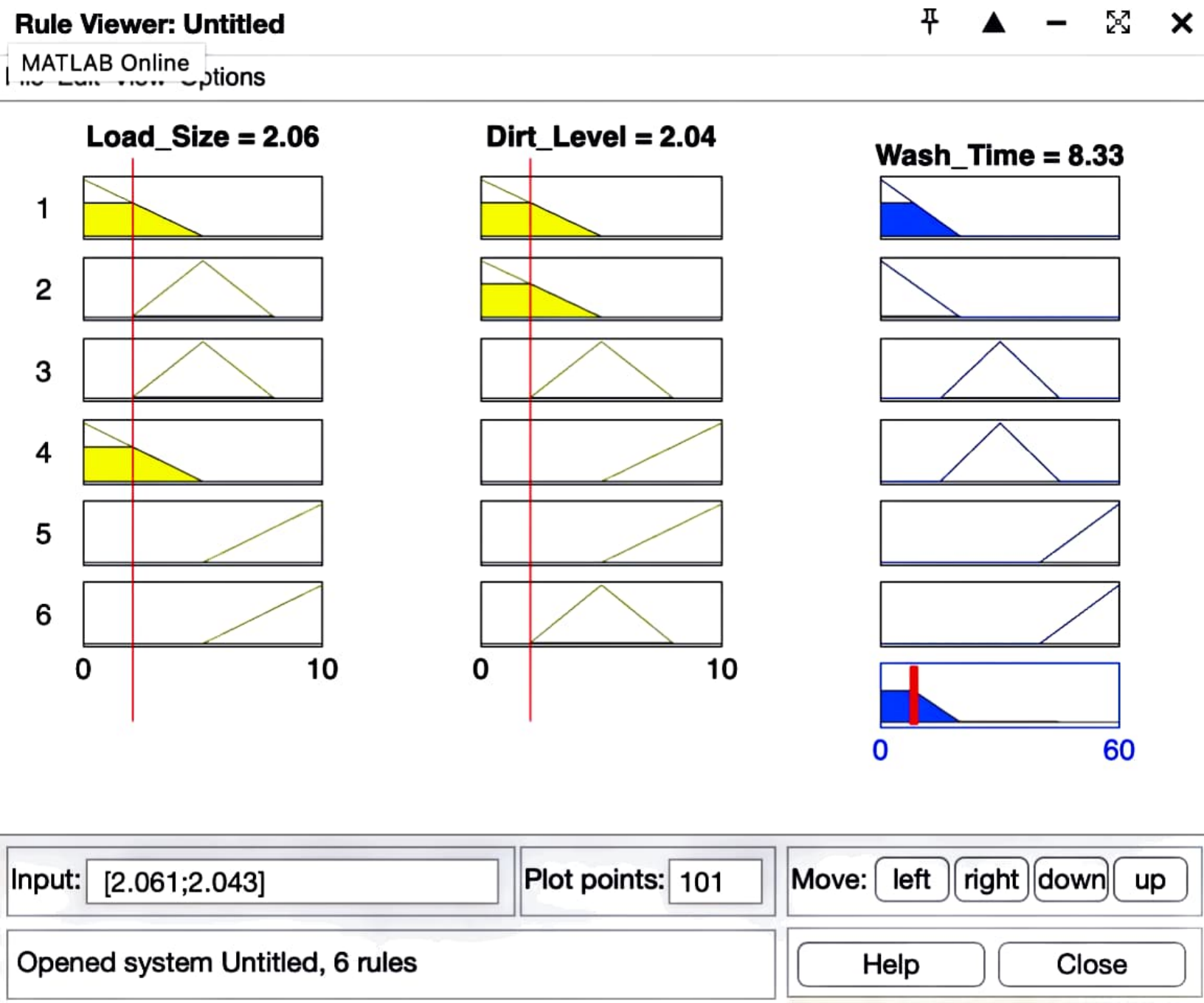
Selected variable "Dirt_Level"

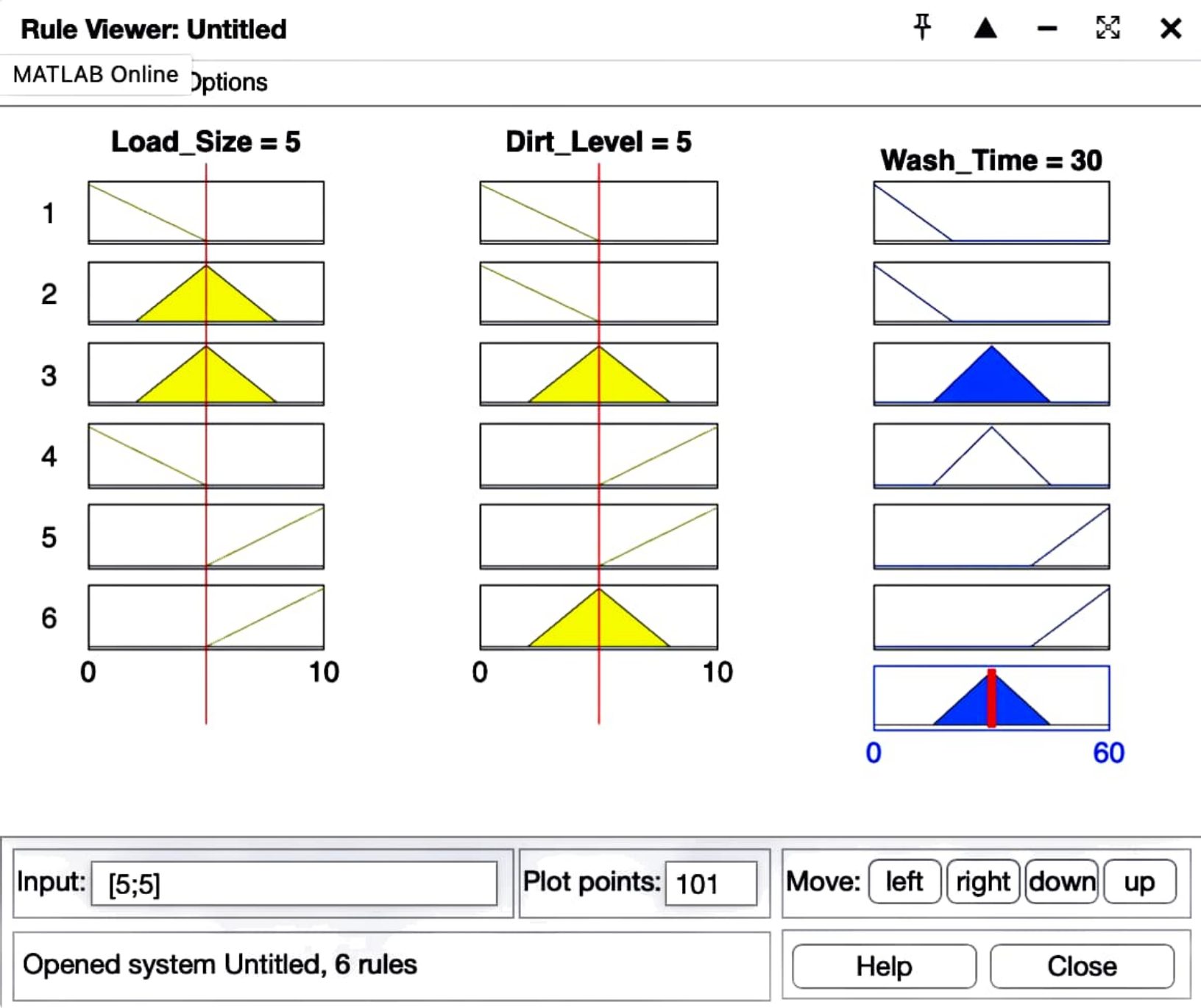

Membership Function Editor: Untitled
MATLAB Online
FIS Variables
Load_Size
Wash_Time
Dirt_Level
Membership function plots
181
Short
Medium
Long
1
0.5
0
0
10
20
30
40
50
60
output variable "Wash_Time"
Current Variable
Name
Wash_Time
Type
output
Range
[0 60]
Display Range
[0 60]
Current Membership Function (click on MF to select)
Name
Short
Type
trimf
Params
[0 0 20]
Help
Close
Selected variable "Wash_Time"

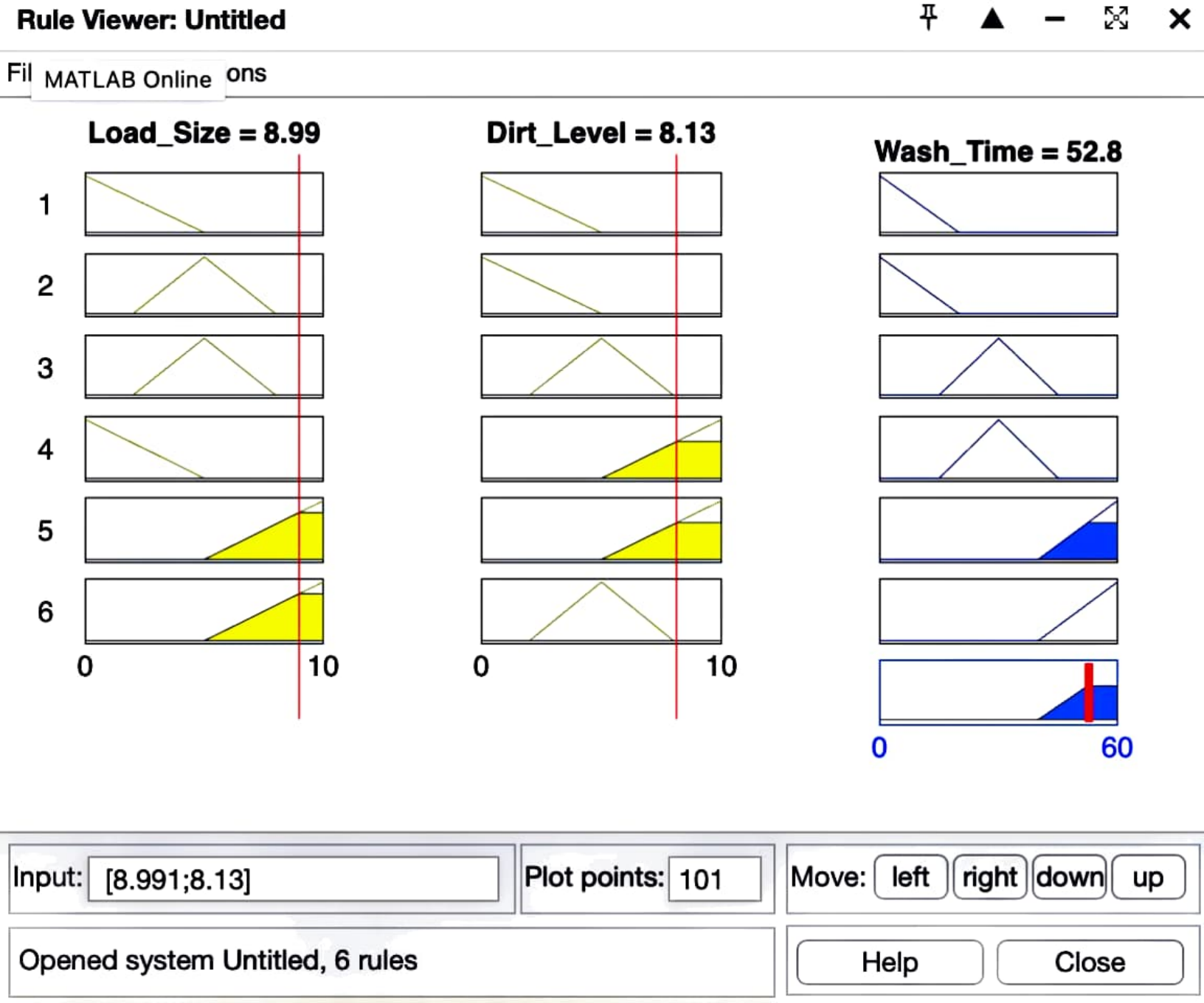

Membership Function Editor: Untitled
MATLAB Online
FIS Variables
Load_Size
Wash_Time
Dirt_Level
Membership function plots
plot points:
181
Low
Medium
High
1
0.5
0
0
1
2
3
4
5
6
7
8
9
10
input variable "Load_Size"
Current Variable
Name
Load_Size
Type
input
Range
[0 10]
Display Range
[0 10]
Current Membership Function (click on MF to select)
Name
Low
Type
trimf
Params
[0 0 5]
Help
Close
Ready


Fuzzy Logic Designer: Untitled
File Edit View
MATLAB Online
Load_Size
Dirt_Level
Untitled
(mamdani)
Wash_Time
FIS Name: Untitled
FIS Type: mamdani
And method min
Or method max
Implication min
Aggregation max
Defuzzification centroid
Current Variable
Name
Type
Range
Help
Close
Ready


Rule Viewer: Untitled
MATLAB Online
Options
Load_Size = 4.96
Dirt_Level = 5
Wash_Time = 30
1
2
3
4
5
6
0
10
0
10
0
60
Input: [4.956;5]
Plot points: 101
Move: left right down up
Opened system Untitled, 6 rules
Help
Close


Rule Viewer: Untitled
MATLAB Online
Load_Size = 2.06
Dirt_Level = 2.04
Wash_Time = 8.33
1
2
3
4
5
6
0
10
0
10
0
60
Input: [2.061;2.043]
Plot points: 101
Move: left right down up
Opened system Untitled, 6 rules
Help
Close


Rule Viewer: Untitled
MATLAB Online
Options
Load_Size = 5
Dirt_Level = 5
Wash_Time = 30
1
2
3
4
5
6
0
10
0
10
0
60
Input: [5;5]
Plot points: 101
Move: left right down up
Opened system Untitled, 6 rules
Help
Close


Rule Viewer: Untitled
Fil
MATLAB Online
ons
Load_Size = 8.99
Dirt_Level = 8.13
Wash_Time = 52.8
1
2
3
4
5
6
0
10
0
10
0
60
Input:
[8.991;8.13]
Plot points:
101
Move:
left
right
down
up
Opened system Untitled, 6 rules
Help
Close